

Machine Learning (ML) and Deep Learning (DL) Notes

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1. Machine Learning (ML)

Machine Learning is a subset of Artificial Intelligence that enables systems to learn from data and improve performance without being explicitly programmed.

Types of Machine Learning

1. Supervised Learning

- Uses labeled data.
- Examples: Linear Regression, Logistic Regression, SVM, Decision Trees.

2. Unsupervised Learning

- Uses unlabeled data.
- Examples: K-Means Clustering, PCA.

3. Reinforcement Learning

- Agent learns through rewards and penalties.
- Examples: Game-playing AI, Robotics.

ML Workflow

- Data Collection
- Data Cleaning
- Feature Engineering
- Model Training
- Evaluation
- Deployment

Common Algorithms

- Linear Regression
- Logistic Regression
- Decision Trees
- Random Forest
- KNN
- SVM
- Naive Bayes

Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1 Score

- ROC-AUC
- Mean Squared Error (MSE)

Advantages

- Easier to train than deep learning.
- Works well with structured data.
- Requires less computational power.

Limitations

- Performance depends on feature engineering.
- Struggles with highly complex data.

2. Deep Learning (DL)

Deep Learning is a subset of Machine Learning based on Artificial Neural Networks with multiple hidden layers.

Neural Network Components

- Input Layer
- Hidden Layers
- Output Layer
- Weights and Biases
- Activation Functions

Popular Activation Functions

- ReLU
- Sigmoid
- Tanh
- Softmax

Types of Deep Learning Networks

1. ANN (Artificial Neural Networks)
2. CNN (Convolutional Neural Networks)
3. RNN (Recurrent Neural Networks)
4. LSTM
5. Transformer Models

Deep Learning Workflow

- Data Collection
- Data Preprocessing
- Model Design
- Training
- Validation
- Testing
- Deployment

Advantages

- Excellent for images, audio, and text.
- Learns features automatically.
- High accuracy on large datasets.

Limitations

- Requires large datasets.
- Computationally expensive.
- Less interpretable.

ML vs DL Comparison

Aspect	ML	DL
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Data Requirement	Small to Medium	Large
Feature Engineering	Manual	Automatic
Training Time	Faster	Slower
Hardware	CPU	GPU/TPU
Interpretability	Higher	Lower
Best For	Structured Data	Images, Audio, NLP

Applications

ML

- Spam Detection
- Recommendation Systems
- Fraud Detection
- Sales Prediction

DL

- Image Recognition
- Speech Recognition
- Chatbots
- Autonomous Vehicles

Important Interview Questions

1. Difference between AI, ML, and DL.
2. Bias-Variance Tradeoff.
3. Overfitting vs Underfitting.
4. What is Gradient Descent?
5. What are CNNs and RNNs?
6. What is Backpropagation?
7. Explain Precision and Recall.